

Ankit Das

Software Engineer | AI/ML

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Summary

Software Engineer building AI/ML-enabled products across multimodal inference, computer vision, full-stack web systems, and native applications. Experienced in taking systems from architecture and evaluation through APIs, security, testing, and deployment.

Experience

Computer Vision Engineer

Feb 2025 – Aug 2026

IDVerse(RELX)

Remote

- Designed document-understanding pipelines across 10+ document types, combining OCR, layout/geometry signals, VLM/LLM inference, candidate resolution, structured output, and API integration.
- Fine-tuned LLM/VLM components with LoRA and QLoRA; built RAG, LangChain, llama.cpp, and schema-constrained generation workflows for controlled experimentation.
- Built FastAPI and Docker services with concurrency-aware processing; benchmarked varied document sizes for sub-minute responses while balancing extraction quality, latency, model footprint, and CPU/browser constraints.
- Applied 4-bit quantization and ONNX/runtime optimization; developed failure analysis across OCR, candidate generation, resolver, and inference behavior.

Quantum Machine Learning Intern

Oct 2023 – Mar 2024

CybraneX

Remote

- Integrated Stripe into Django and contributed to TrOCR experimentation and an Azure-based ML pipeline.

Selected Projects

UnScroll – Multimodal Visual Extraction Engine

TypeScript, Bun, Next.js, VLM APIs, Zod

- Architected a typed monorepo with CLI, REST API, and Next.js interfaces; implemented bounded-concurrency VLM extraction, canonical-link resolution, structured Markdown exports, and provider-neutral adapters.
- Added schema/MIME validation, SHA-256 digests, explicit pipeline states, SSRF defenses, and verified 42 automated tests.
- Kept model providers, storage adapters, and pipeline states replaceable behind typed contracts with explicit retry and failure paths.

APU Matrix – Mobile Price-Performance Platform

Next.js, React, Data Pipelines, Vitest

- Built a tested dashboard and data pipeline joining phone prices with separate CPU, GPU, AI, and overall benchmark families while preserving provenance, compatibility metadata, and explicit missingness; verified 134 automated tests.
- Implemented processor normalization, device matching, currency and price-selection rules, API-backed views, accessible charts, responsive tables, filtering, sorting, and pagination.

Pune Metro Video Analytics – Queue Monitoring Prototype

Python, YOLOv8, OpenCV, Object Tracking

- Implemented detection, persistent tracking, region-based queue counting, threshold alerts, annotated video output, and custom metro-footage training experiments.
- Separated detection, tracking, counting, alerting, and visualization stages to support iterative validation of video-analytics behavior.

Photo Memories – Local macOS Photo Widget

Swift, SwiftUI, WidgetKit, AppIntents

- Built a privacy-conscious macOS app/widget using security-scoped bookmarks, App Groups, bounded thumbnail timelines, and persistent local references without copying user photos.

Technical Skills

Languages/Web: Python, SQL, TypeScript, JavaScript, Swift, Next.js, React, Bun, FastAPI, Django, Flask, REST APIs, gRPC

AI/ML/RAG: PyTorch, TensorFlow, OpenCV, Hugging Face Transformers, OCR/KIE, VLMs, LLM/VLM fine-tuning, LoRA/QLoRA, RAG, hybrid retrieval, Qdrant, BM25, knowledge graphs, LangChain, LangGraph

Retrieval/Evaluation: Query rewriting and decomposition, cross-encoder reranking, evidence verification, citation grounding, provenance tracking, retrieval/generation evaluation, ablation experiments, structured outputs

Inference/Platform: ONNX Runtime, llama.cpp, model export, 4-bit quantization, preprocessing/post-processing, benchmarking, profiling, concurrency, Docker, PostgreSQL, GitHub Actions, GCP, Microsoft Azure

Quality/Product: Zod, Vitest, XCTest, automated testing, schema validation, API security, SSRF/MIME defenses, system design, failure analysis

Education and Leadership

B.Tech, Computer Engineering with Honours in Data Science | 2021 – 2025

Shah & Anchor Kutchhi Engineering College, Mumbai | CGPA: 8.05/10

Grand Finalist, Smart India Hackathon 2023 | **President and Administration Lead**, SAKEC Research Cell

Website and App Secretary, Computer Society of India – SAKEC Student Chapter